- *Update APIs*—To edit and modify the fields of a document
- *Delete APIs*—To remove documents from the store

Elasticsearch classifies these APIs into two categories: single-document APIs and multi-document APIs. As the name suggests, *single-document APIs* perform operations like indexing, fetching, modifying, and deleting documents one by one, using the appropriate endpoints. These APIs are useful when working with events such as orders generated by an e-commerce application or tweets from a set of Twitter handles. We use single-document APIs to operate on these documents individually.

Multi-document APIs, on the other hand, are geared toward working with documents in batches. For example, we may have a requirement to import a product catalog into Elasticsearch from a database consisting of millions of records. Elasticsearch exposes a bulk (`_bulk`) endpoint API for this purpose to help import data in batches.

Elasticsearch also exposes advanced query-based APIs to work with our documents. We can delete and update a number of documents that match certain criteria by developing a complex query if needed. We can also use sophisticated search queries to find and update or delete documents. Finally, we can move data from one index to another using reindexing APIs. Reindexing helps us migrate our data without any downtime in production, although we must consider the performance implications.

In this chapter, we discuss single- and multi-document APIs and various operations on documents. The first step in working with Elasticsearch is to get some data into the engine. Let's begin by looking at the indexing document APIs and the mechanics of indexing documents.

NOTE The code for this chapter is available on GitHub (http://mng.bz/MBr8) and on the book's website (https://www.manning.com/books/elasticsearch-in-action-second-edition). You can find the movie data set used in the examples at http://mng.bz/a1pX.

5.1 Indexing documents

Just as we insert records into a relational database, we add data (in the form of documents) to Elasticsearch. These documents sit in a logical bucket called an *index*. The act of persisting documents to these indexes is known as *indexing*. So, when we hear the term *indexing*, it means storing or persisting documents into Elasticsearch.

Text analysis

The documents undergo a process called *text analysis* in Elasticsearch before being stored. The analysis process prepares data to make it suitable for various search and analytical features. This analysis gives the search engine the ability to deliver relevancy and full-text search capabilities. Text analysis is discussed in depth in chapter 7.